

CUSTOMER SHOPPING BEHAVIOUR ANALYSIS REPORT

A Data Analytics Project using Python, SQL & Power BI

Source Repository:

github.com/PRASANNA-163/Customer_Behaviour_Analysis

Dataset: Retail_Customer_Insights.csv (3,900 customer records, 18 attributes)

Prepared: July 2026

1. Executive Summary

This report presents an end-to-end data analytics case study built on the Customer_Behaviour_Analysis GitHub repository. The project analyses 3,900 retail transactions across 18 attributes (demographics, product category, purchase amount, discounts, subscriptions, ratings, and shipping) using a three-stage pipeline: Python for data preparation, PostgreSQL for analytical querying, and Power BI for interactive dashboarding.

The repository's own Business Problem Document frames the objective as helping a retail company “leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies.” This report reconstructs that pipeline, re-executes every SQL query in the repository against the source dataset, reports the actual query outputs, and reproduces the Power BI dashboard's KPIs and visuals so the findings can be reviewed in a single document.

Headline numbers

Metric	Value
Total customers analysed	3,900
Average purchase amount	\$59.76
Average review rating	3.75 / 5
Total revenue (all customers)	\$233,081
Male vs Female revenue share	68% vs 32%
Customers with an active subscription	27% (1,053)

2. Business Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. Management has noticed changes in purchasing patterns across demographics, product categories, and sales channels, and is particularly interested in which factors — discounts, reviews, seasons, or payment preferences — drive consumer decisions and repeat purchases.

Overarching business question:

Insight: “How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?”

Project deliverables defined in scope

- Data Preparation & Modeling (Python) — clean and transform the raw dataset.
- Data Analysis (SQL) — structure the data and run queries on customer segments, loyalty, and purchase drivers.
- Visualization & Insights (Power BI) — build an interactive dashboard for stakeholders.
- Report & Presentation — summarize key findings and business recommendations.
- GitHub Repository — host all Python scripts, SQL queries, and dashboard files.

3. Dataset Overview

The analysis is built on `Retail_Customer_Insights.csv`, containing 3,900 rows and 18 original columns describing individual retail purchase transactions.

Column	Description
Customer ID	Unique identifier for each customer
Age / Gender	Customer demographics
Item Purchased / Category	Product bought and its category (Clothing, Footwear, Outerwear, Accessories)
Purchase Amount (USD)	Transaction value
Location, Size, Color, Season	Purchase context attributes
Review Rating	Customer's product rating (1–5)
Subscription Status	Whether the customer is a subscriber
Shipping Type	Standard, Express, Free Shipping, etc.
Discount Applied / Promo Code Used	Whether a discount/promo was used on the order
Previous Purchases	Count of prior purchases by the customer
Payment Method / Frequency of Purchases	How and how often the customer buys

4. Data Preparation — Python

The notebook “`Customer_Shopping_Behaviour_Analysis.ipynb`” uses pandas to load, inspect, and clean the raw CSV before loading it into PostgreSQL for SQL analysis. The key transformation steps found in the notebook are:

4.1 Initial inspection

- Loaded the CSV with `pandas.read_csv()` and reviewed `df.head()`, `df.info()`, `df.describe(include="all")`, and `df.isnull().sum()` — the dataset was found to have no missing values.

4.2 Feature engineering — Age Group

Customers were bucketed into four equal-sized quantile groups using `pd.qcut()` on Age:

```
labels = ['Young Adult', 'Adult', 'Middle-Aged', 'Senior']
df['Age Group'] = pd.qcut(df['Age'], q=4, labels=labels)
```

4.3 Feature engineering — Purchase Frequency in Days

The categorical “Frequency of Purchases” field was mapped to a numeric day-count so it can be aggregated and compared quantitatively:

```
FrequencyMapping = {
    'Fortnightly':14, 'Weekly':7, 'Monthly':30, 'Quarterly':90,
    'Bi-Weekly':14, 'Annually':365, 'Every 3 Months':90
}
```

```
}  
df['Purchase Frequency Days'] = df['Frequency of Purchases'].map(FrequencyMapping)
```

4.4 Redundancy removal

The notebook checks whether Discount Applied and Promo Code Used are identical across every row — `(df['Discount Applied'] == df['Promo Code Used']).all()` returns True — confirming the two columns are redundant. Promo Code Used is then dropped to avoid duplicated signal.

4.5 Column standardization & load to PostgreSQL

Column names are lower-cased and spaced replaced with underscores (e.g. “Purchase Amount (USD)” → `purchase_amount`) so they map cleanly to SQL identifiers. The cleaned DataFrame is then pushed into a PostgreSQL database (`customer_behaviour`) via SQLAlchemy's `df.to_sql("customer", engine, if_exists="replace")` — this is the customer table referenced throughout the SQL analysis in Section 5.

5. SQL Analysis & Query Results

All 10 business questions defined in `Customer_Behaviour_Analysis.sql` were re-executed against the cleaned dataset (using DuckDB as a PostgreSQL-equivalent SQL engine over the same customer table) so the actual results could be reported here.

Q1. What is the total revenue generated by male vs. female customers?

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer GROUP BY gender;
```

Gender	Total Revenue (USD)
Male	\$157,890
Female	\$75,191

Insight: Male customers generate roughly 2.1x the revenue of female customers (\$157,890 vs \$75,191), driven primarily by a larger share of male transactions in the dataset rather than a materially higher average order value.

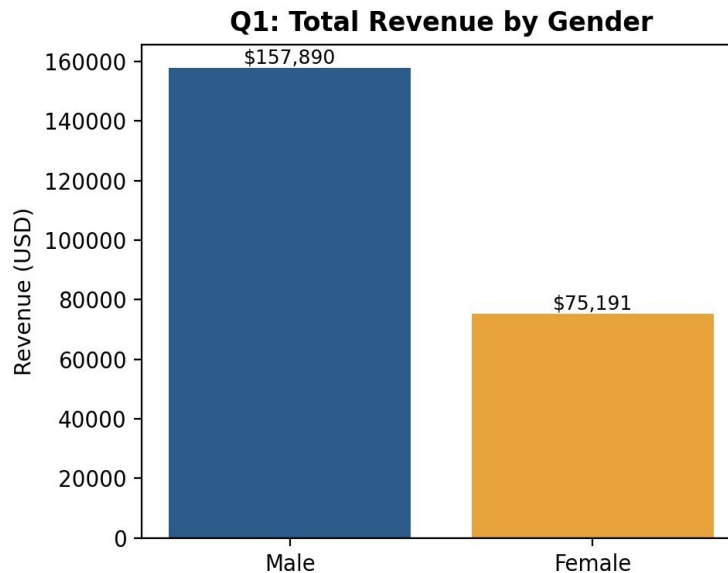


Figure 5.1 — Total revenue by gender

Q2. Which customers used a discount but still spent more than the average purchase amount?

```
SELECT customer_id, purchase_amount FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

Result summary	Value
Customers matching this profile	839
Share of all discount-users	42.9% of the 1,955 customers who used a discount
Average spend within this group	\$79.90 (vs overall average of \$59.76)

Result summary	Value
Spend range	\$60 – \$100

Insight: 839 customers (about 1 in 5 of the full base) used a discount and still spent above the \$59.76 average — evidence that discounts are not purely driving down-market behaviour; a meaningful segment of higher-value shoppers uses discounts opportunistically on top of larger baskets.

Q3. Which are the top 5 products with the highest average review rating?

```
SELECT item_purchased, ROUND(AVG(review_rating),2) AS avg_rating
FROM customer GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC LIMIT 5;
```

Rank	Item Purchased	Avg. Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.81
4	Hat	3.81
5	T-shirt	3.78

Insight: Ratings are tightly clustered between 3.78 and 3.86 for the top 5 items — there is no single stand-out “best-loved” product; accessories and footwear items (Gloves, Sandals, Boots) edge out apparel on satisfaction.

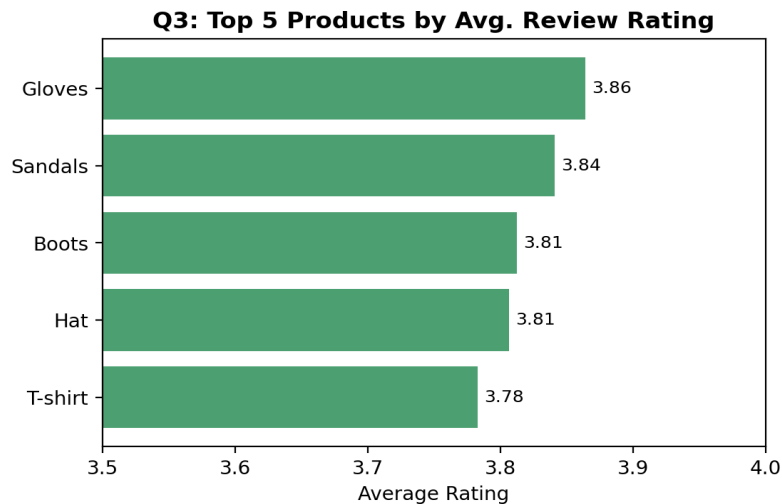


Figure 5.2 — Top 5 products by average review rating

Q4. Compare the average purchase amount between Standard and Express shipping.

```
SELECT shipping_type, ROUND(AVG(purchase_amount),2)
FROM customer WHERE shipping_type IN ('Standard','Express')
GROUP BY shipping_type;
```

Shipping Type	Avg. Purchase Amount (USD)
Express	\$60.48

Shipping Type	Avg. Purchase Amount (USD)
Standard	\$58.46

Insight: Customers who choose Express shipping spend marginally more on average (\$60.48 vs \$58.46) — about a 3.5% premium — suggesting a weak but positive link between order value and willingness to pay for faster delivery.

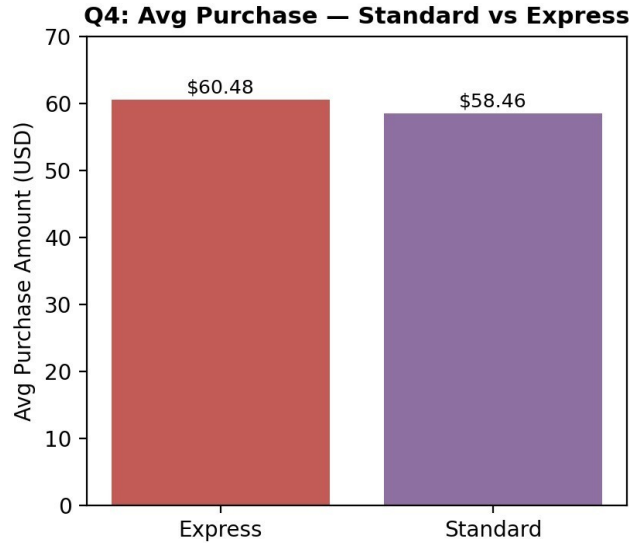


Figure 5.3 — Average purchase amount: Standard vs Express shipping

Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.

```
SELECT subscription_status, COUNT(customer_id) AS total_customer,
       ROUND(AVG(purchase_amount), 2) AS avg_spend,
       ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer GROUP BY subscription_status
ORDER BY total_revenue DESC;
```

Subscription Status	Customers	Avg. Spend	Total Revenue
No	2,847	\$59.87	\$170,436
Yes	1,053	\$59.49	\$62,645

Insight: Subscribers and non-subscribers spend almost identically per order (\$59.49 vs \$59.87) — subscription status is not a meaningful driver of basket size. Non-subscribers generate the majority of total revenue simply because they are ~2.7x more numerous (2,847 vs 1,053), pointing to a large opportunity to convert high-value non-subscribers into subscribers rather than relying on subscription to lift spend.

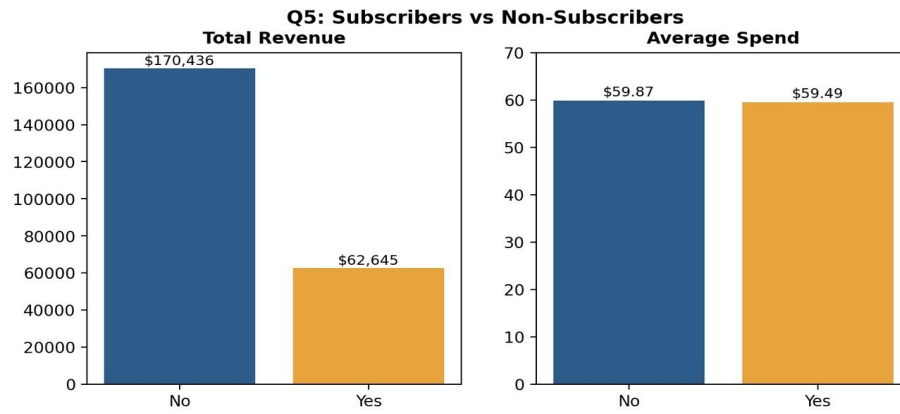


Figure 5.4 — Subscribers vs non-subscribers: total revenue and average spend

Q6. Which 5 products have the highest percentage of purchases with discounts applied?

```
SELECT item_purchased,
       ROUND(100*SUM(CASE WHEN discount_applied='Yes' THEN 1 ELSE 0 END)/COUNT(*),2) AS
       discount_rate
FROM customer GROUP BY item_purchased
ORDER BY discount_rate DESC LIMIT 5;
```

Rank	Item Purchased	Discount Rate
1	Hat	50.0%
2	Sneakers	49.7%
3	Coat	49.1%
4	Sweater	48.2%
5	Pants	47.4%

Insight: Discount usage is fairly uniform across the top items (47–50%), with Hats leading at an even 50%. No single product category is being discounted disproportionately, suggesting discount codes are applied broadly rather than targeted at slow-moving inventory.

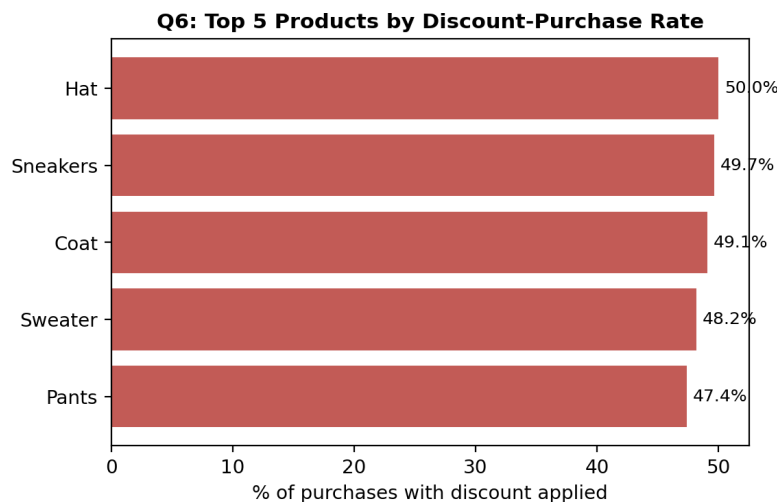


Figure 5.5 — Top 5 products by discount-purchase rate

Q7. Segment customers into New, Returning and Loyal based on their total number of previous purchases, and show the count of each segment.

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE WHEN previous_purchases = 1 THEN 'New'
         WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
         ELSE 'Loyal' END AS customer_segment
  FROM customer)
SELECT customer_segment, COUNT(*) AS num_customers
FROM customer_type GROUP BY customer_segment;
```

Segment	Definition	Number of Customers
Loyal	>10 previous purchases	3,116 (79.9%)
Returning	2–10 previous purchases	701 (18.0%)
New	1 previous purchase	83 (2.1%)

Insight: The customer base is dominated by Loyal customers (nearly 80%), which is unusually high for a retail file and suggests the sample already skews toward an established, repeat-purchasing customer base rather than reflecting a typical new-customer acquisition funnel. Retention programs should therefore focus on deepening engagement with this large Loyal cohort while a dedicated onboarding push targets the small New segment.

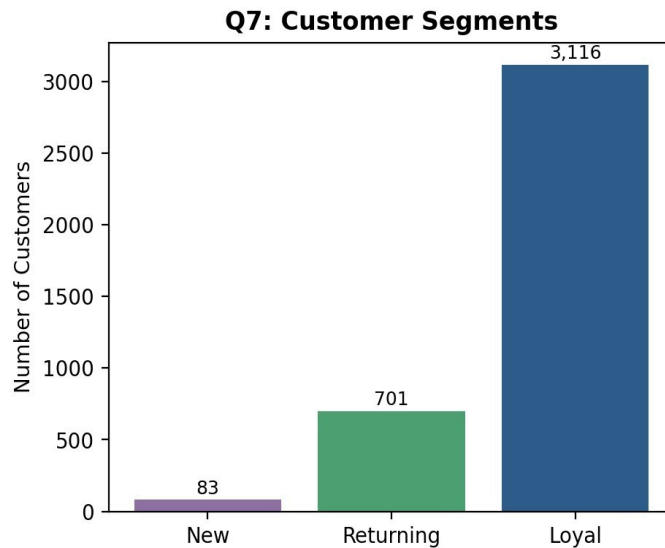


Figure 5.6 — Customer segments: New, Returning, Loyal

Q8. What are the 3 most purchased products within each category?

```
WITH item_counts AS (
  SELECT category, item_purchased, COUNT(customer_id) AS total_orders,
    ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank
  FROM customer GROUP BY category, item_purchased)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts WHERE item_rank <= 3;
```

Category	#1	#2	#3
Clothing	Pants (171)	Blouse (171)	Shirt (169)

Category	#1	#2	#3
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	— (only 2 items in category)
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)

Insight: Order volumes are remarkably even within each category (typically within 10–15 orders of one another), meaning no single product dominates demand — assortment breadth, not a hero SKU, is driving category-level sales. The Outerwear category has only two products in the catalogue (Jacket, Coat), so it naturally has no third-ranked item.

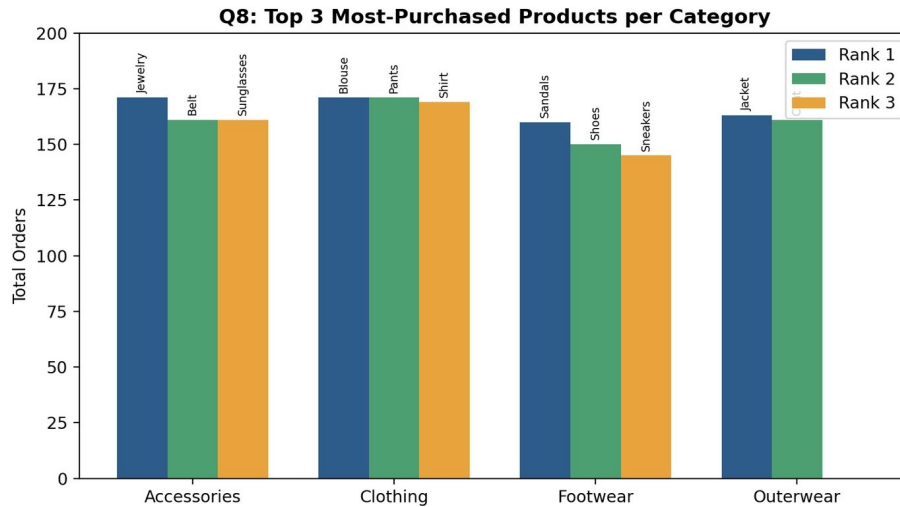


Figure 5.7 — Top 3 most-purchased products per category

Q9. Are repeat buyers (more than 5 previous purchases) also likely to subscribe?

```
SELECT subscription_status, COUNT(customer_id) AS repeat_buyers
FROM customer WHERE previous_purchases > 5
GROUP BY subscription_status;
```

Subscription Status	Repeat Buyers (>5 previous purchases)
No	2,518 (72.4%)
Yes	958 (27.6%)

Insight: Among repeat buyers, the subscribed share (27.6%) is essentially the same as the overall subscription rate (27.0%) — being a repeat buyer does not meaningfully increase the likelihood of subscribing. Subscription conversion appears to be independent of purchase frequency, so subscription marketing should not assume loyal/repeat customers are already “halfway converted.”

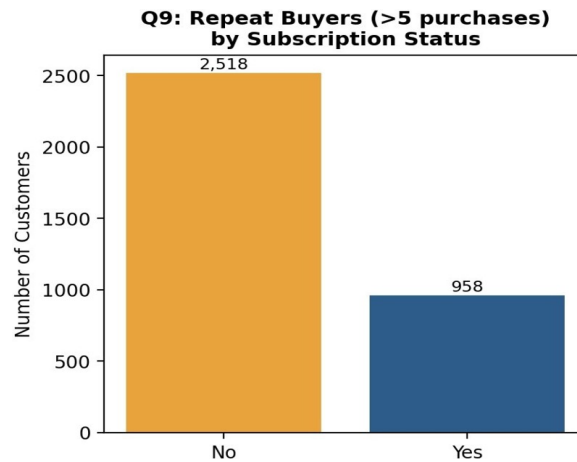


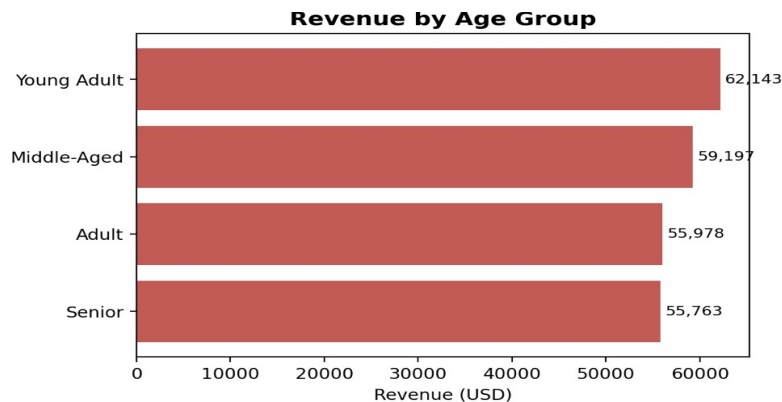
Figure 5.8 — Repeat buyers (>5 previous purchases) by subscription status

Q10. What is the revenue contribution of each age group?

```
SELECT age_group, SUM(purchase_amount) AS total_revenue
FROM customer GROUP BY age_group
ORDER BY total_revenue DESC;
```

Age Group	Total Revenue (USD)
Young Adult	\$62,143
Middle-Aged	\$59,197
Adult	\$55,978
Senior	\$55,763

Insight: Revenue is spread very evenly across the four (quantile-balanced) age groups, with Young Adults contributing slightly the most (\$62,143) and Seniors slightly the least (\$55,763) — a spread of only about 11% between the highest and lowest group. Age is not a strong differentiator of spend, implying marketing spend need not be heavily skewed toward any one age bracket.



6. Power BI Dashboard



The repository includes Customer_Behaviour_Dashboard.pbix, a single-page interactive Power BI report titled “Customer Behaviour Dashboard.”

6.1 Dashboard structure (as defined in the .pbix report layout)

Visual	Type	Fields / Measures
KPI Card	cardVisual	Number of Customers, Average Purchase Amount, Average Review Rating
% of Customer by Subscription Status	Donut chart	subscription_status, Sum(customer_id)
Revenue by Category	Clustered column chart	category, Sum(purchase_amount)
Sales by Category	Clustered column chart	category, Sum(customer_id)
Revenue by Age Group	Clustered bar chart	age_group, Sum(purchase_amount)
Sales by Age Group	Clustered bar chart	age_group, Sum(customer_id)
Slicers	Advanced slicer	Subscription Status, Gender, Category

The three slicers (Subscription Status, Gender, Category) let a stakeholder filter every visual on the page interactively — e.g. isolating revenue-by-category for female subscribers only.

6.2 KPI Cards

KPI Card	Value
Number of Customers	3,900
Average Purchase Amount	\$59.76
Average Review Rating	3.75

6.3 % of Customers by Subscription Status

% of Customers by Subscription Status

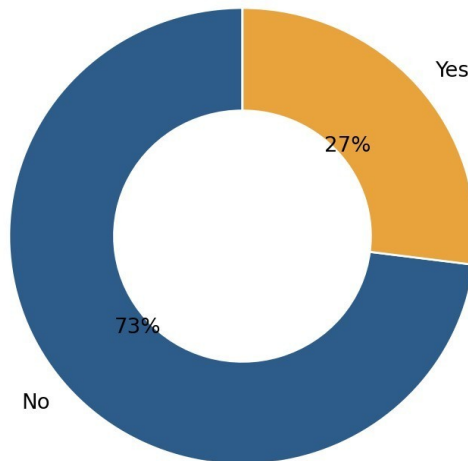


Figure 6.1 — Reconstructed donut chart: 73% not subscribed, 27% subscribed

6.4 Revenue and Sales by Category

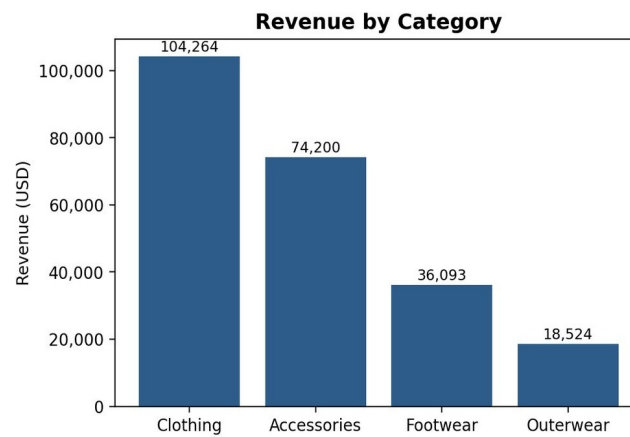


Figure 6.2 — Revenue by category

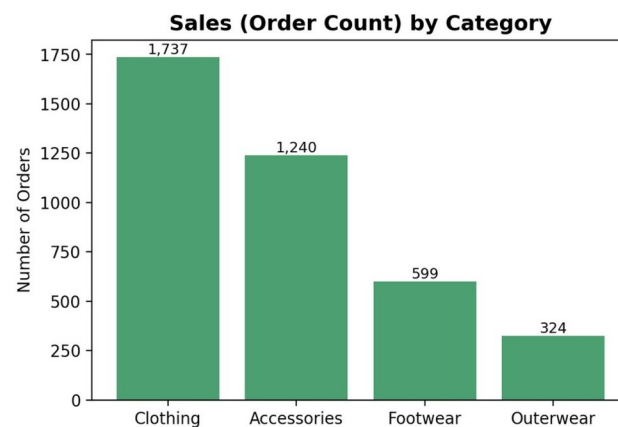


Figure 6.3 — Sales (order count) by category

Insight: Clothing leads on both revenue (\$104,264) and order count (1,737 orders), followed by Accessories. Outerwear is the smallest category on both dimensions, reflecting its narrower product range (only Jacket and Coat).

6.5 Revenue and Sales by Age Group

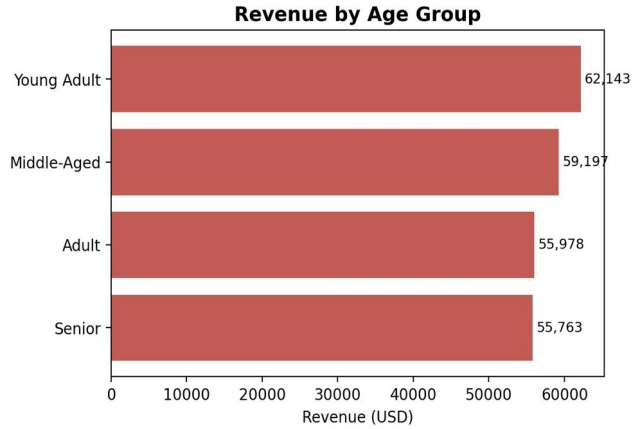


Figure 6.4 — Revenue by age group

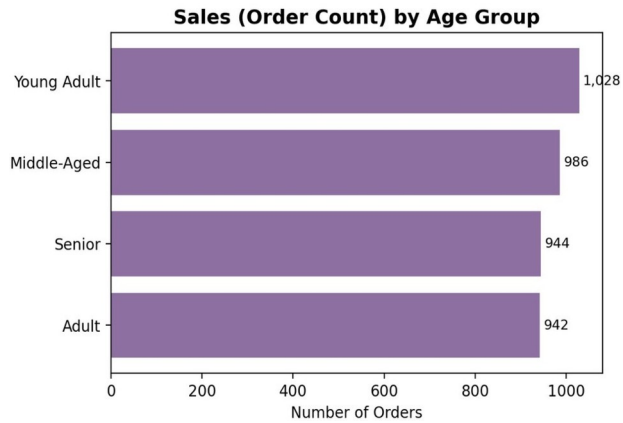


Figure 6.5 — Sales (order count) by age group

Insight: Order counts and revenue by age group track each other closely and are all within a narrow band, confirming (as in Q10) that age is not a strong lever for targeting revenue growth on its own.

7. Key Insights Summary

- Male customers drive over two-thirds of total revenue (\$157,890 vs \$75,191), largely due to transaction volume rather than higher basket size.
- A meaningful segment (839 customers, ~22% of the base) uses discounts while still spending above average — discounts are not purely a value-shopper tool.
- Product ratings are tightly clustered (3.75–3.86 average); no product is a runaway favourite or a clear laggard.
- Express shipping customers spend only ~3.5% more than Standard shipping customers — shipping speed is a weak proxy for order value.
- Subscription status has almost no effect on per-order spend (\$59.49 vs \$59.87); the subscriber base is simply smaller (27%), so it contributes less total revenue.
- Discount usage is broad-based across products (47–50% of purchases for the top 5 items) rather than concentrated on a few SKUs.
- ~80% of customers already qualify as “Loyal” (>10 previous purchases) — the dataset represents an already-retained customer base.
- Category demand is broad-based: the top 3 products within each category are within 10–15 orders of each other, indicating no single hero product.
- Being a frequent repeat buyer does not increase the likelihood of subscribing (27.6% vs the 27.0% base rate) — subscription conversion is independent of purchase frequency.
- Revenue is evenly spread across age groups (a max spread of only ~11% between the highest and lowest group), so age alone is not a strong targeting variable.

8. Business Recommendations

- Since subscription does not lift order value, reposition subscription marketing around retention/convenience benefits (free shipping, early access) rather than “spend more” messaging, and target the large non-subscriber base (73%) directly rather than assuming loyal buyers will self-convert.
- Investigate the demand drivers behind the higher male revenue share (channel mix, category preference, or marketing spend allocation) to decide whether to grow male revenue further or invest in closing the gender gap.
- Use the discount-but-above-average-spend segment (839 customers) as a testbed for premium bundling or upsell offers layered on top of standard discounts, since this group has already shown willingness to spend more.
- Broaden the Outerwear catalogue (currently only Jacket and Coat) given it is the smallest category on both revenue and volume — assortment expansion may be a faster lever than pricing changes.
- Since age and shipping type are weak differentiators of spend, prioritize other segmentation variables (e.g. purchase frequency, category preference, discount sensitivity) for targeted campaigns instead.
- Because ~80% of customers are already “Loyal,” build a formal loyalty/rewards tier to protect and further monetize this base, while running a separate, low-cost onboarding flow for the small New-customer segment.

9. Technical Stack & Repository Contents

File	Purpose
Customer_Shopping_Behaviour_Analysis.ipynb	Python/pandas notebook: data cleaning, feature engineering, and load to PostgreSQL
Customer_Behaviour_Analysis.sql	10 business-question SQL queries against the cleaned `customer` table
Customer_Behaviour_Dashboard.pbix	Power BI interactive dashboard (KPIs, donut chart, category & age-group charts, slicers)
Retail_Customer_Insights.csv	Source dataset (3,900 rows x 18 columns)
Business Problem Document.pdf	Business problem statement and project deliverables
README.md / LICENSE	Project description (MIT license)

10. Conclusion

The Customer_Behaviour_Analysis project delivers a complete, reproducible analytics workflow — from raw CSV to a governed SQL layer to an interactive Power BI dashboard — that directly answers the business's stated question about identifying trends and optimizing marketing and product strategy. The re-executed queries confirm that revenue is concentrated by gender and by an already-loyal customer base, while age, shipping type, and subscription status are comparatively weak differentiators of individual spend. These findings give the business concrete, low-ambiguity levers (subscription repositioning, assortment expansion in Outerwear, and loyalty-program investment) to act on.